

HDOC Day-29

C Program: Menu-Driven Array Operations

Question: Write a C program to count even, odd, positive, negative and zero elements in an array; search an element using linear search; merge two arrays; and exit using a menu-driven program.

Program

```
#include <stdio.h>

int main()
{
    int choice, n, i, key, found;
    int a[100], b[100], c[200];
    int n1, n2;
    int even, odd, positive, negative, zero;

    do
    {
        printf("\n--- MENU ---\n");
        printf("1. Count even, odd, positive, negative and zero\n");
        printf("2. Linear search\n");
        printf("3. Merge two arrays\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1:
                printf("Enter number of elements: ");
                scanf("%d", &n);

                printf("Enter %d elements:\n", n);
                for(i = 0; i < n; i++)
                    scanf("%d", &a[i]);

                even = odd = positive = negative = zero = 0;

                for(i = 0; i < n; i++)
                {
                    if(a[i] % 2 == 0)
                        even++;
                    else
                        odd++;

                    if(a[i] > 0)
                        positive++;
                    else if(a[i] < 0)
                        negative++;
                    else
                        zero++;
                }

                printf("Even elements = %d\n", even);
                printf("Odd elements = %d\n", odd);
                printf("Positive elements = %d\n", positive);
                printf("Negative elements = %d\n", negative);
                printf("Zero elements = %d\n", zero);
                break;

            case 2:
                printf("Enter number of elements: ");
                scanf("%d", &n);

                printf("Enter %d elements:\n", n);
                for(i = 0; i < n; i++)
```

```

        scanf("%d", &a[i]);

printf("Enter element to search: ");
scanf("%d", &key);

found = 0;
for(i = 0; i < n; i++)
{
    if(a[i] == key)
    {
        printf("Element found at position %d\n", i + 1);
        found = 1;
        break;
    }
}

if(found == 0)
    printf("Element not found\n");
break;

case 3:
    printf("Enter size of first array: ");
    scanf("%d", &n1);
    printf("Enter elements of first array:\n");
    for(i = 0; i < n1; i++)
        scanf("%d", &a[i]);

    printf("Enter size of second array: ");
    scanf("%d", &n2);
    printf("Enter elements of second array:\n");
    for(i = 0; i < n2; i++)
        scanf("%d", &b[i]);

    for(i = 0; i < n1; i++)
        c[i] = a[i];

    for(i = 0; i < n2; i++)
        c[n1 + i] = b[i];

    printf("Merged array:\n");
    for(i = 0; i < n1 + n2; i++)
        printf("%d ", c[i]);
    printf("\n");
    break;

case 4:
    printf("Program terminated.\n");
    break;

default:
    printf("Invalid choice. Please try again.\n");
}

} while(choice != 4);

return 0;
}

```

Short Explanation

- Option 1:** Reads an array and counts even, odd, positive, negative and zero elements.
- Option 2:** Uses linear search to find a given element and displays its position.
- Option 3:** Reads two arrays and combines them into a third array.
- Option 4:** Ends the program.

The **do-while** loop keeps displaying the menu until the user chooses 4.